Sampling and Sampling Distributions

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Objectives

- Explain why we sample, and distinguish probability from nonprobability samples.

- Draw and correctly analyse simple random, systematic, stratified, and cluster samples, and say what each design buys.
- Calculate sampling errors, and identify nonsampling errors that no sample size can fix.
- Build a sampling distribution of the mean, and use its mean μ and its standard error $\sigma/\sqrt{n}$.
- State the Central Limit Theorem with its conditions, and recognize when a sample is not yet large enough.
- Find probabilities for a sample mean, and the interval that contains 95% of sample means.
- Use a sampling distribution to judge a claim, and describe the result in the right words.
- Use the sampling distribution of the proportion, with a stated policy on the continuity correction.
- Use the finite population correction, and say why it exists.
- Use the bootstrap to estimate a standard error from the one sample you have.

Chapters 5 and 6 described data generating mechanisms: if you know the mechanism, what values will it produce? This chapter asks the question a manager actually faces, which runs the other way. She has one sample, and from it one number: an average order size, a share of satisfied customers. How far might that number be from the truth she cannot see?

The answer is another probability distribution. Draw a different sample and you get a different average. The distribution of all the averages you could have gotten is the **sampling distribution**, and its standard deviation, the **standard error**, is the single most important number in the second half of this course. Every confidence interval in Chapter 8 and every hypothesis test after it is built from one.

As before, you will not compute these by hand on an exam. You will be handed output and asked what it means. In this chapter the most important thing it can mean is "this sample is (or is not) the kind of sample that claim would produce."

Why Sample?

A **population** is all possible subjects of interest in a study. A **sample** is a subset of the population. A **census** records every member of the population.

Nielsen estimates the viewing habits of roughly 129 million US television households from a panel of about 41,000 of them. A cracker maker checks salt content by testing crackers, and a tested cracker cannot be sold. A university that wants students' views on academic integrity could survey all of them, but would spend far more to learn very little more.

Those are the three reasons to sample: a census would be **impossible** (the population is too large or keeps changing), **destructive** (measuring uses up the item), or **wasteful** (a well-drawn sample answers the question almost as well, for a fraction of the cost). The price of sampling is that the answer is uncertain. The point of this chapter is that the uncertainty can be measured.

Types of Sampling

A **probability sample** is one in which every member of the population has a known, nonzero chance of being selected. The chances need not be equal.

A **nonprobability sample** is one in which the chance of selection is not known.

Only a probability sample supports the calculations in the rest of this chapter. If you do not know how members got into the sample, you cannot say how far the sample is likely to be from the population.

Throughout this section the population is real: the 215 economics faculty members at nine Big Ten universities in 1999, with their salaries (Baser and Pema, 2003). Because we have the whole population, we know the true mean salary, \$93,457, and can see how well each design recovers it.

rank	faculty	mean salary	sd
Assistant	20	58974.0	8655.0
Associate	40	68722.0	12067.0
Professor	155	104289.0	26913.0

Simple random sampling

> In a **simple random sample** every member of the population has the same chance of being chosen, and every possible sample of size n is equally likely. Members are drawn **without replacement**: nobody is chosen twice.

```
Sample of 30: mean salary $94,905,  population mean $93,457,  sampling error $1,449
```

	id	institution	rank	sex	salary
1	102	Indiana	Professor	Male	100450
10	111	Indiana	Professor	Male	96470
24	206	Iowa	Professor	Male	66594
26	208	Iowa	Associate	Female	81250
43	303	Michigan State	Professor	Male	113307
53	314	Michigan State	Professor	Male	94915
55	316	Michigan State	Associate	Male	59772
58	319	Michigan State	Professor	Female	66512

> Excel's **Data Analysis → Sampling** tool draws *with* replacement, so it can hand you the same member twice. To draw a simple random sample in Excel, put `=RAND()` in a column beside the list, sort on that column, and take the first n rows.

Systematic sampling

> In **systematic sampling**, every kth member of a list is chosen, starting from a random position among the first k. The **systematic sampling constant** is
>
> $$k = \frac{N}{n},$$
>
> where N is the population size and n the sample size.

Systematic sampling is easy to carry out by hand: every fifth customer through the door, every hundredth file in the cabinet. It behaves like a simple random sample as long as the list has no pattern that lines up with k.

> **Periodicity** is a pattern in the list that repeats with the same spacing as k. It can make a systematic sample unrepresentative.

Here is periodicity on purpose. A Summerlin restaurant's daily covers over a year have a weekly cycle, quiet on weekdays and busy on weekends. (The numbers are an illustration, not data.) Sample every 7th day and you sample the *same weekday* every time:

```
True mean daily covers: 268.3
```

design first day	k = 5 (every 5th day)	k = 7 (every 7th day)
Monday	267.4	204.9
Tuesday	263.7	201.1
Wednesday	272.7	216.4
Thursday	266.3	233.9
Friday	270.1	343.5
Saturday	(no such start)	378.6
Sunday	(no such start)	299.7

With $k = 7$, the answer depends entirely on the starting day: from 201 covers if you start on a quiet weekday to 379 if you start on a Saturday, against a true mean of 268. With $k = 5$, the sample cycles through every weekday and all five possible samples land between 264 and 273. Before using systematic sampling, ask what order the list is in.

Stratified sampling

> In a **stratified sample**, the population is divided into mutually exclusive groups, or **strata**, and a random sample is taken from each. The strata are chosen because the variable of interest differs across them.

Faculty salaries differ sharply by rank, so rank is a natural stratum. There are two ways to set the stratum sample sizes.

Proportional allocation makes the sample mirror the population. For a sample of 30:

stratum	population count	share of population	n x share	sample size
Professor	155	0.721	21.628	22
Associate	40	0.186	5.581	5
Assistant	20	0.093	2.791	3

With proportional allocation the ordinary sample mean is already the right estimate, because each stratum appears in its population share.

Disproportionate allocation samples some strata more heavily, usually to learn enough about a small group. With only 20 assistant professors, proportional allocation gives us 3 of them, too few to say anything about assistant professors. Sampling 10 from each rank fixes that, but now the sample over-represents the lower-paid ranks, and its plain average is biased.

The **stratified estimate** of the population mean weights each *stratum mean* by that stratum's share of the population, $W_h = N_h/N$:

$$\bar{x}_{\text{strat}} = \sum_h W_h \, \bar{x}_h.$$

Equivalently, each sampled observation from stratum h gets weight W_h/n_h.

stratum	N_h	W_h = N_h / N	n_h	stratum sample mean	W_h x mean
Professor	155	0.721	10	98757.1	71196.979
Associate	40	0.186	10	63645.4	11841.005
Assistant	20	0.093	10	57849.0	5381.302

The plain average of these 30 salaries is \$73,417; the stratified estimate is \$88,419; the truth is \$93,457. One sample proves little, so here is each design run 1,000 times:

```
True mean salary: $93,457
```

design	average estimate	bias	standard error	RMSE
simple random sample, n = 30	93441.0	−15.0	4925.0	4923.0
stratified, proportional (3/5/22)	93914.0	458.0	3754.0	3780.0
stratified 10/10/10, plain average	77208.0	−16248.0	3009.0	16524.0
stratified 10/10/10, weighted by W_h	93453.0	−3.0	6004.0	6001.0
stratified 6/9/15, each observation weighted W_h (wrong)	97632.0	4175.0	5290.0	6737.0

Read the `bias` and `standard error` columns together. **Bias** is how far the design's estimates sit from the truth *on average*; the **standard error** is how much they scatter from sample to sample. RMSE combines the two.

- Proportional stratification beats a simple random sample of the same size: a standard error of \$3,754 against \$4,925. Making sure every rank is represented in its proper share removes the luck of drawing too many or too few full professors. That is what stratification is for.
- The 10/10/10 sample's plain average is too low by \$16,248 on average. It is not a bad sample; it is a bad *estimate*, because it ignores how the sample was drawn. Weighting by W_h removes the bias.
- Weighting each *observation* by W_h, instead of each *stratum mean*, is a tempting mistake. With unequal stratum sample sizes it leaves a bias of \$4,175, because it counts the heavily sampled strata twice: once through n_h and again through W_h.

Cluster sampling

In **cluster sampling**, the population is divided into mutually exclusive groups, or **clusters**, each of which is ideally a small version of the whole population. A random sample of *clusters* is chosen, and everyone (or a random sample) within the chosen clusters is measured. Clusters are often geographic.

The nine universities are natural clusters. A researcher who must travel to collect data could visit **3 of the 9 campuses**, chosen at random, and record every economist there. Note what that is *not*: visiting all nine campuses and sampling within each is *stratified* sampling, with campuses as strata.

```
Campuses drawn: Illinois, Indiana, Purdue;  61 faculty;  mean $88,253
```

design	average estimate	bias	standard error	RMSE
cluster: 3 of 9 campuses, everyone	93675.0	218.0	4077.0	4081.0
simple random sample, n = 72	93650.0	193.0	2968.0	2973.0

Three random campuses average about 72 faculty, so the comparison is with a simple random sample of 72. The cluster sample's standard error is $4,077 against $2,968. Members of a cluster resemble each other (Michigan pays more than Purdue), so 72 economists from three campuses carry less information than 72 drawn from all nine.

Stratified versus cluster. The two are easy to confuse, because both divide the population into groups.

	Stratified	Cluster
Groups are	internally alike (all full professors)	mixed, each a mini-population
You sample	from **every** group	**some** groups, entirely
Purpose	more **accuracy**	lower **cost**
Precision vs. a simple random sample of the same size	usually better	usually worse

A cluster sample is the right choice when the cost saved (three trips instead of nine) buys a larger sample than the precision lost.

Nonprobability sampling

A **convenience sample** is one whose members are chosen because they are easy to reach.

Customers who happen to walk in, your own students, an online poll, people who choose to write a review: all are convenience samples. They are fine for a quick reaction to a new product. They do not support the probability calculations in the rest of this chapter, because nobody knows how members got in.

A bigger convenience sample is not a better one. In 1936 the *Literary Digest* mailed ballots to names drawn from telephone directories and automobile registrations, people who in the Depression were disproportionately well off and Republican. About 2.4 million responded, and the *Digest* predicted that Alf Landon would beat Franklin Roosevelt. Roosevelt won in a landslide. Gallup, using a far smaller sample chosen to represent the electorate, called the winner.

The error had nothing to do with sample size and everything to do with *who* could be sampled. Twelve years later, in 1948, the major polls predicted Thomas Dewey would beat Harry Truman, in part because they used quota samples (interviewers filled demographic quotas with whomever they chose) and stopped polling weeks before an electorate that was still moving.

Identify the sampling technique in each case.

(a) The first Monday of each month, customers who come into a store are asked to complete a satisfaction survey.

(b) Four stores in a mall are chosen at random, and every customer in those four stores is asked about a new parking policy.

(c) An interviewer stands at a busy corner on Fremont Street and asks passers-by about the Golden Knights.

(d) 60% of a college's students are women. A random sample is built that is 60% women, drawn at random from the women and from the men separately.

(e) Software picks 20 employees at random from the payroll file for a job satisfaction survey.

i Solution

(a) Systematic (a fixed interval in time), with a periodicity worry: first Mondays may not be typical days. **(b)** Cluster. **(c)** Convenience. **(d)** Stratified, with proportional allocation. **(e)** Simple random.

Sampling and Nonsampling Errors

A **parameter** is a number that describes a population, such as its mean μ. A **statistic** is a number calculated from a sample, such as the sample mean $\bar{x}$. Statistics are used to estimate parameters.

The **sampling error** of the sample mean is the difference between the statistic and the parameter:

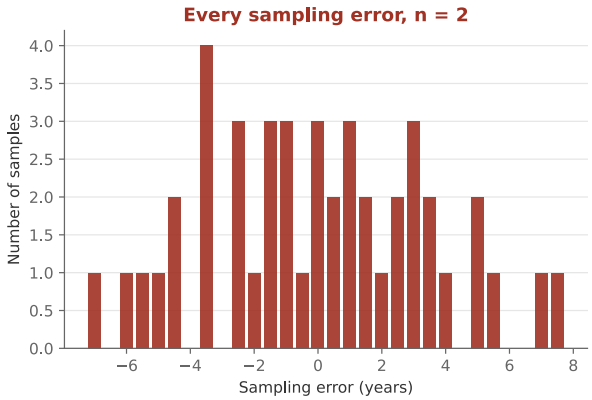
$$\text{sampling error} = \bar{x} - \mu.$$

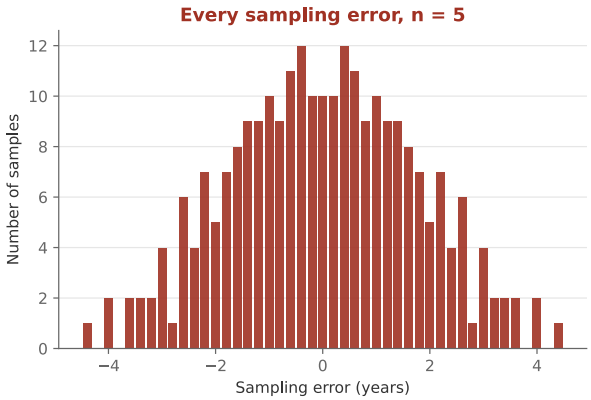
A small population makes sampling error something you can see completely. A Henderson marketing office has ten employees, with these ages:

```
Ages: [24, 33, 28, 41, 38, 29, 26, 33, 37, 31]    mu = 32.0
```

Take the first two employees as a sample: $\bar{x} = (24 + 33)/2 = 28.5$, a sampling error of $28.5 - 32.0 = -3.5$ years. The last two give $(37 + 31)/2 = 34.0$, an error of $+2.0$. But there is no need to pick samples one at a time. With ten people there are only 45 possible samples of two and 252 samples of five, and we can look at all of them:

	possible samples	mean of the sample means	share with |error| <= 2 years	largest negative error	largest positive error
n = 2	45	32.0	0.422	−7.0	7.5
n = 5	252	32.0	0.754	−4.4	4.4





Three lessons, each visible in the table:

1. **Sampling errors vary from sample to sample, and can be positive or negative.** Which one you get is luck.
2. **They average out.** The mean of all possible sample means is exactly $\mu = 32.0$, for both sample sizes.
3. **Larger samples make small errors more likely, not certain.** 42% of samples of two land within two years of μ; 75% of samples of five do. But the unluckiest sample of five, {33, 41, 38, 33, 37}, is still off by +4.4 years.

In practice μ is unknown, so we never know the sampling error of the sample we actually drew. What we can know, as the rest of this chapter shows, is how *large* sampling errors tend to be.

Nonsampling errors

Nonsampling errors arise from how data are collected rather than from which members happened to be sampled: ambiguous or leading questions, a sampling frame that leaves people out, people who choose whether to respond, recording mistakes, faulty instruments.

Sampling error shrinks as the sample grows. **Nonsampling error does not.** Asking a biased question of 10,000 people gives a more precise estimate of the wrong thing.

Application: what online reviews can and cannot tell you

A product has 25,596 online ratings: 88% five stars, 8% four, 2% three, 1% two, and 1% one. Treating these as a sample of customers, the average and its standard error are easy to compute:

```
25,596 ratings: mean = 4.810, s = 0.612, standard error s/sqrt(n) = 0.0038
```

With a standard error of a few thousandths of a star, the sampling error is negligible. The real question is whether reviewers are a random sample of customers, and they are not: people choose whether to review, and the delighted and the furious are the most likely to bother. Suppose, for illustration, that among *all* customers the ratings would be spread 35/30/15/10/10% from five stars to one, and that a five-star customer writes a review 25% of the time, a one-star customer 15%, and everyone else 3–5%:

reviews	mean of reviews	standard error s/sqrt(n)	population mean	bias
100	4.260	0.137	3.7	0.560

reviews	mean of reviews	standard error s/sqrt(n)	population mean	bias
1000	4.182	0.044	3.7	0.482
10000	4.209	0.014	3.7	0.509

As the number of reviews grows from 100 to 10,000, the standard error falls to 0.014 and the bias stays at about +0.51 stars. A thousand times more reviews buys a thousand times more confidence in a number that is still wrong.

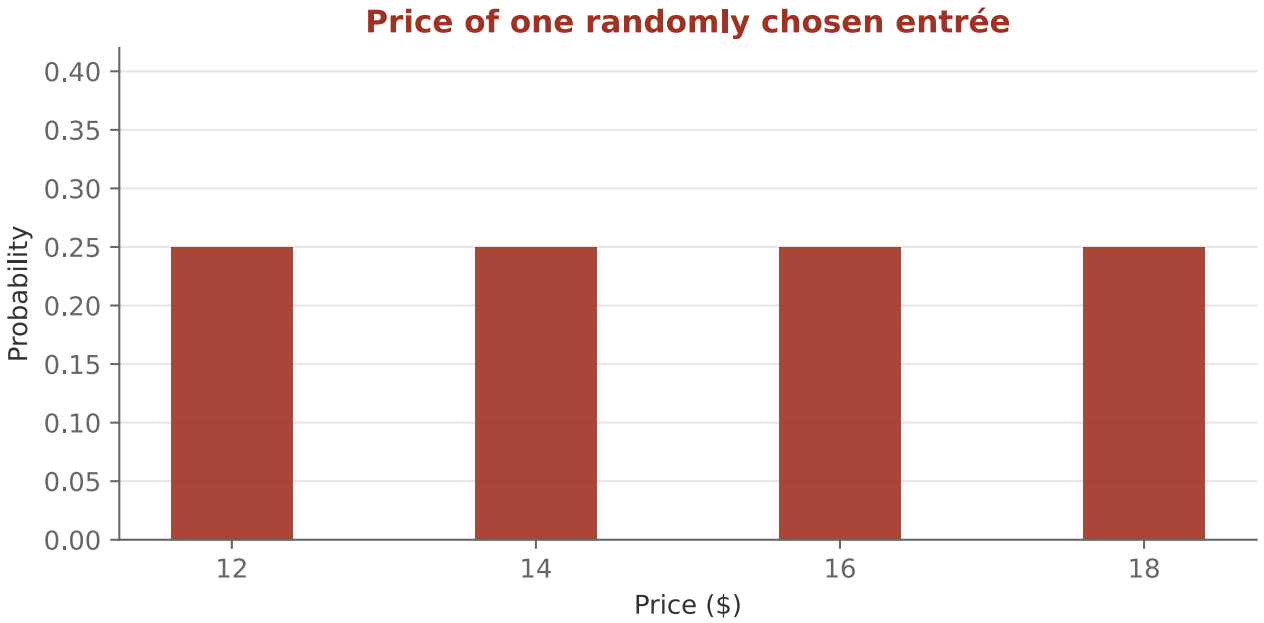
> One more caution about that example. Star ratings are **ordinal** data: the gap between two and three stars need not equal the gap between four and five. Averaging them treats them as interval data. It is universal practice, and your textbook does it too, but know that you are doing it.

The Sampling Distribution of the Mean

Every possible sample

A restaurant offers four entrées: vegetarian $12, chicken $14, fish $16, and steak $18. A diner picks one at random, so the price is a discrete distribution with each value equally likely.

```
Population: mu = 15.00, sigma = 2.2361
```



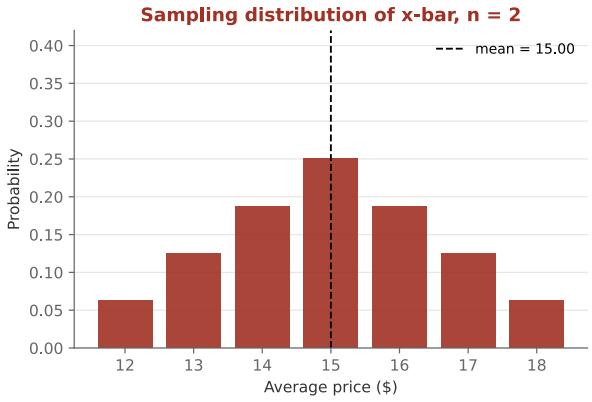
You and a friend each choose an entrée at random. That is a random sample of size $n = 2$ drawn *with replacement* (you may both choose fish). There are $4 \times 4 = 16$ equally likely samples. Here is the average price for every one of them, and the resulting distribution of the sample mean:

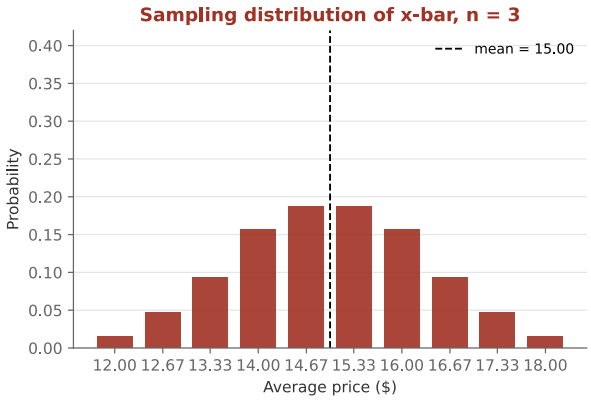
```
Average price for each of the 16 samples:

           friend: $12  friend: $14  friend: $16  friend: $18
you: $12         12.0         13.0         14.0         15.0
```

```
you: $14        13.0        14.0        15.0        16.0
you: $16        14.0        15.0        16.0        17.0
you: $18        15.0        16.0        17.0        18.0
```

x-bar	P(x-bar)	number of samples
12.0	0.0625	1.0
13.0	0.1250	2.0
14.0	0.1875	3.0
15.0	0.2500	4.0
16.0	0.1875	3.0
17.0	0.1250	2.0
18.0	0.0625	1.0





The **sampling distribution of the mean** is the probability distribution of $\bar{x}$ over all possible samples of size n from a population.

Look at what happened. The population is flat: every price equally likely. The sampling distribution of $\bar{x}$ for two diners is already peaked at $15, because an average of $15 can happen four ways and an average of $12 only one. With three diners (64 samples), it is starting to look like a bell. Averaging pulls values toward the middle. That is the Central Limit Theorem beginning to work, and we will meet it formally in a moment.

Now compute the mean and standard deviation of the sampling distribution, exactly as for any discrete distribution in Chapter 5:

	population (n = 1)	x-bar, n = 2	x-bar, n = 3
mean	15.0000	15.0000	15.000
standard deviation	2.2361	1.5811	1.291
sigma / sqrt(n)	NaN	1.5811	1.291

The mean of the sample means equals the population mean, $15, every time. The standard deviation of the sample means shrinks, and it equals $\sigma/\sqrt{n}$ exactly: 2.2361 divided by $\sqrt{2}$ is 1.5811.

For samples of size n from a population with mean μ and standard deviation σ:

$$\mu_{\bar{x}} = \mu \qquad \text{and} \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}.$$

$\sigma_{\bar{x}}$ is the **standard error of the mean**: the standard deviation of the sampling distribution of $\bar{x}$. It measures how far a sample mean typically lands from μ.

Two consequences worth saying out loud.

The sample mean is **unbiased**: it is not systematically too high or too low. Any single sample can miss, but the misses average to zero.

The standard error shrinks with the **square root** of n. Quadrupling the sample halves the standard error; to cut it by a factor of ten, you need a hundred times the data. Precision gets expensive, which is why sample size is a business decision and not only a statistical one.

Probabilities for a sample mean

When the sampling distribution of $\bar{x}$ is normal, probabilities for a sample mean come from the z-score of $\bar{x}$:

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

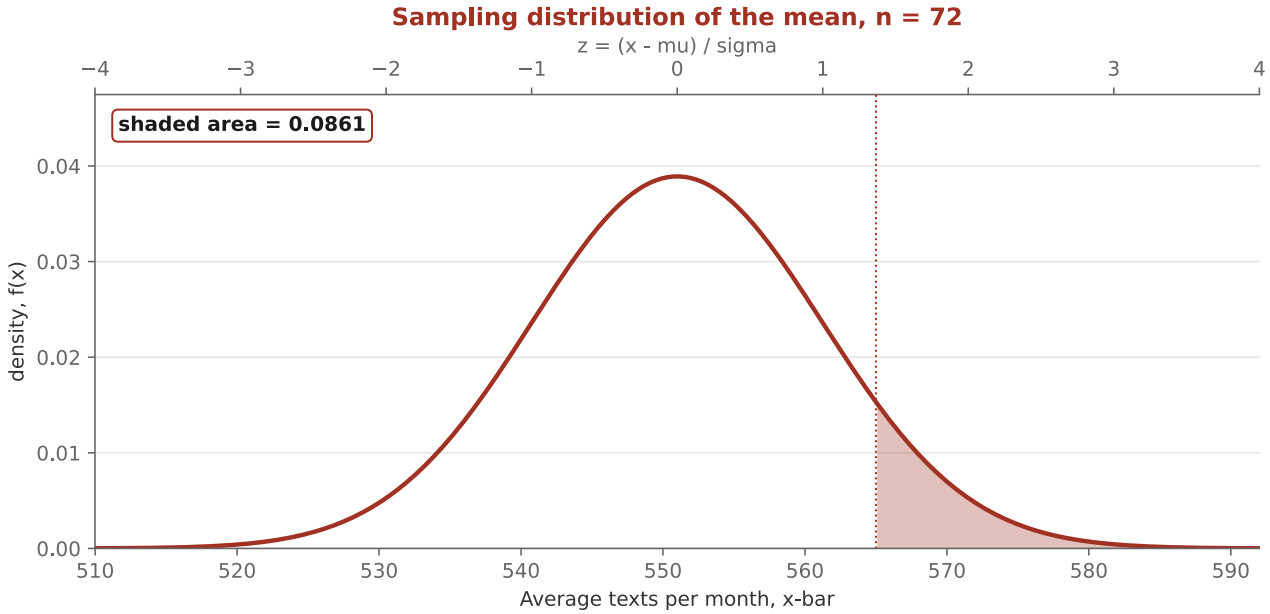
The denominator is the **standard error**, $\sigma/\sqrt{n}$, not σ. Dividing by σ answers a question about *one person*. Dividing by $\sigma/\sqrt{n}$ answers a question about the *average of n people*. This is the most common mistake in the chapter, and it produces an answer that looks perfectly reasonable.

Application: text messages

Americans aged 18 to 54 send and receive an average of 551 text messages a month. Suppose the population standard deviation is 87. What is the probability that a random sample of 72 people averages more than 565?

	72 people
standard error	10.253048
finite population correction	1.0
event (x-bar)	X > 565
shape	right tail
z lower	1.365448
z upper	+inf
P(X <= upper)	1.0
P(X <= lower)	0.913944
computed as	1 - P(X <= 565)
probability	0.086056
Excel	=1 - NORM.DIST(565, 551, 10.253, TRUE)

$$\sigma_{\bar{x}} = \frac{87}{\sqrt{72}} = 10.2530 \qquad z = \frac{565 - 551}{10.2530} = 1.3654 \qquad P(\bar{x} > 565) = 0.0861$$



The probability is about 8.6%. Had we divided by σ instead of the standard error, we would have reported 43.6%, which is the probability that *one person* exceeds 565 (and only if individual texting were normal, which nobody has claimed). Notice that we did not need to know the shape of the population. The next section explains why.

The Central Limit Theorem

The Central Limit Theorem (CLT). For samples of size n drawn from any population with mean μ and standard deviation σ, the sampling distribution of $\bar{x}$ is **approximately normal** with mean μ and standard error $\sigma/\sqrt{n}$, and the approximation improves as n grows.

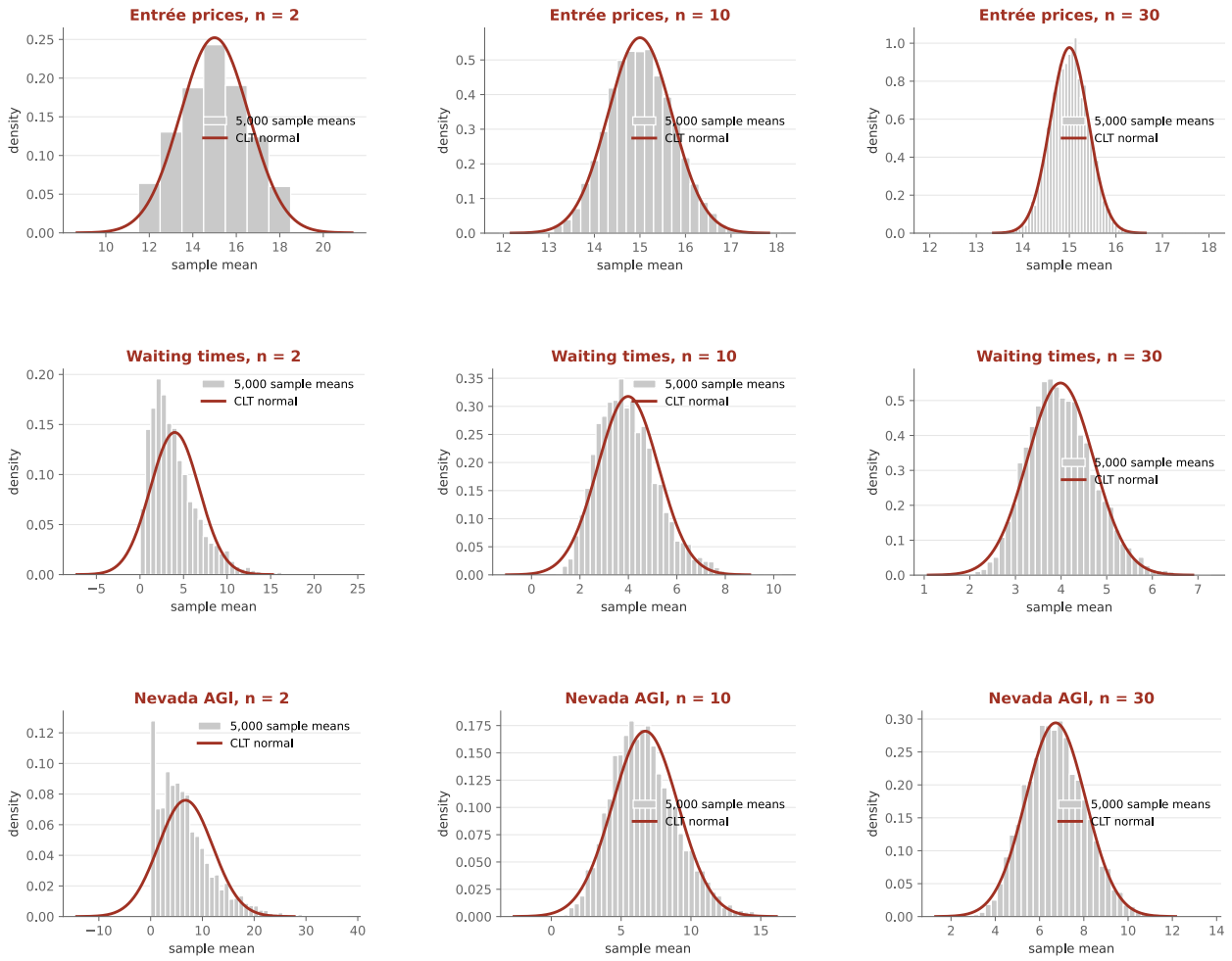
- If the population is itself **normal**, $\bar{x}$ is exactly normal for **any** n.
- Otherwise, $n \geq 30$ is the usual **rule of thumb** for "large enough." The more skewed the population, the larger n must be.

This is the reason statistics is a required course. It says that averages behave predictably even when individuals do not, so a manager can make probability statements about an average without knowing the shape of the population it came from.

Watching it happen

Three populations: a flat one (the entrées), a right-skewed one (exponential waiting times from Chapter 6, mean 4 minutes), and Nevada adjusted gross income, which is strongly right-skewed. For each, we draw 5,000 *fresh* samples of each size from the population and plot the sample means, with the normal curve the CLT predicts on top.

	mean	sd	median
Entrée prices (flat)	15.000	2.236	15.000
Waiting times (exponential)	3.988	3.971	2.775
Nevada AGI (right-skewed)	6.727	7.427	4.808



Read down each column. The flat population's means are close to normal by $n = 10$. The waiting times need more: at $n = 2$ the means are still skewed, and by $n = 30$ the histogram and the curve nearly agree. Nevada income is the hardest case.

When $n = 30$ is not enough

The normal model says 2.5% of sample means fall more than 1.96 standard errors above μ, and 2.5% more than 1.96 below. For Nevada AGI, here is what actually happens:

	below mu - 1.96 SE	above mu + 1.96 SE	normal model, each tail
n = 30	0.0138	0.0300	0.025
n = 100	0.0218	0.0278	0.025
n = 400	0.0250	0.0216	0.025
n = 1600	0.0244	0.0238	0.025

> At $n = 30$, 3.0% of sample means land above the upper cut-off and only 1.4% below the lower one. The normal model says 2.5% each. A skewed population produces a skewed sampling distribution at moderate n: occasional samples catch a few very high incomes and their averages jump up, while no sample can catch incomes below zero. The tails even out by a few hundred.

The practical rule: $n \geq 30$ is a starting point, not a guarantee. For incomes, insurance claims, hospital bills, order sizes, and anything else with a long right tail, look at the data and expect to need more. (The AGI file is simulated from IRS zip-code counts; its shape is real, its rows are not. See `data/README.md`.)

The CLT is about **averages**, not individuals. It says nothing about the probability that *one* Nevada household earns more than a given amount; that depends entirely on the skewed population. It lets you say how the *average of n households* behaves.

How Much Do Sample Means Vary?

The interval that contains 95% of sample means

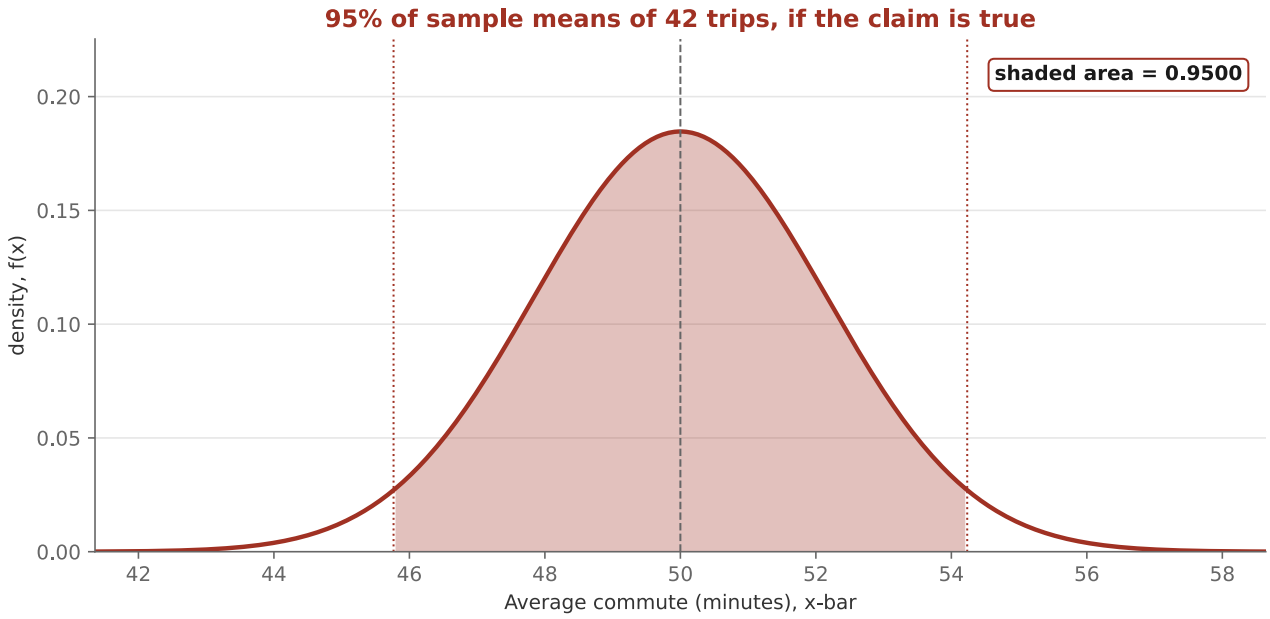
If $\bar{x}$ is normal with mean μ and standard error $\sigma_{\bar{x}}$, then 95% of all sample means fall within

$$\mu \pm 1.96\,\sigma_{\bar{x}} = \mu \pm 1.96\,\frac{\sigma}{\sqrt{n}}.$$

The 1.96 is `NORM.S.INV(0.975)`: it leaves 2.5% in each tail.

A navigation app claims the average morning commute on a route is 50 minutes, with a standard deviation of 14 minutes. If the claim is true, where should the average of 42 trips fall?

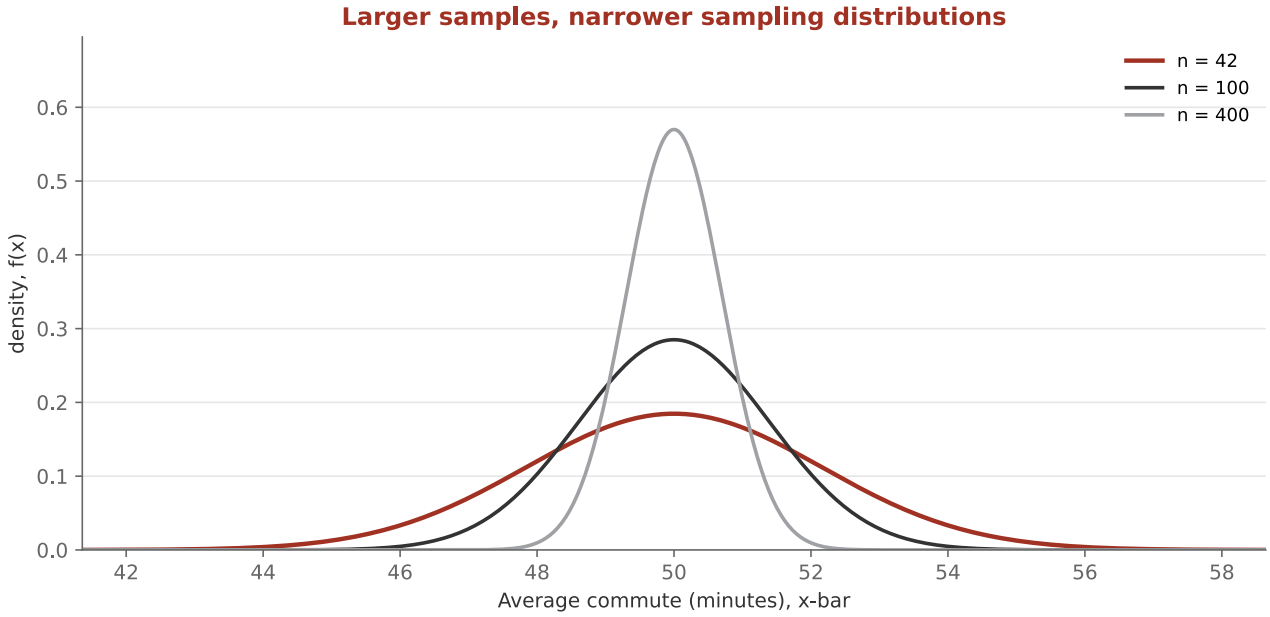
	n = 42
level	0.9500
z	1.9600
standard error	2.1602
margin z x SE	4.2340
lower	45.7660
upper	54.2340



If the claim is true, 95% of samples of 42 trips will average between 45.8 and 54.2 minutes.

The effect of sample size

n	standard error	lower	upper	width
42	2.16	45.77	54.23	8.47
100	1.40	47.26	52.74	5.49
168	1.08	47.88	52.12	4.23
400	0.70	48.63	51.37	2.74



Going from 42 trips to 168, four times the data, cuts the standard error and the interval's width exactly in half. The sampling distribution gets taller and narrower: sample means crowd around μ.

> **Toward Chapter 8.** In practice we do not know μ; we know $\bar{x}$. But if $\bar{x}$ is within $1.96\,\sigma_{\bar{x}}$ of μ in 95% of samples, then μ is within $1.96\,\sigma_{\bar{x}}$ of $\bar{x}$ in 95% of samples. Turn this interval around and it becomes a **confidence interval**, the subject of Chapter 8.

Taking it to the limit: a census

As the sample grows toward the whole population, the sample mean closes in on μ. Measure everyone and $\bar{x} = \mu$: the sampling error is zero. The formula $\sigma/\sqrt{n}$ does not quite know this, because it assumes an infinite population. The fix is in the next section.

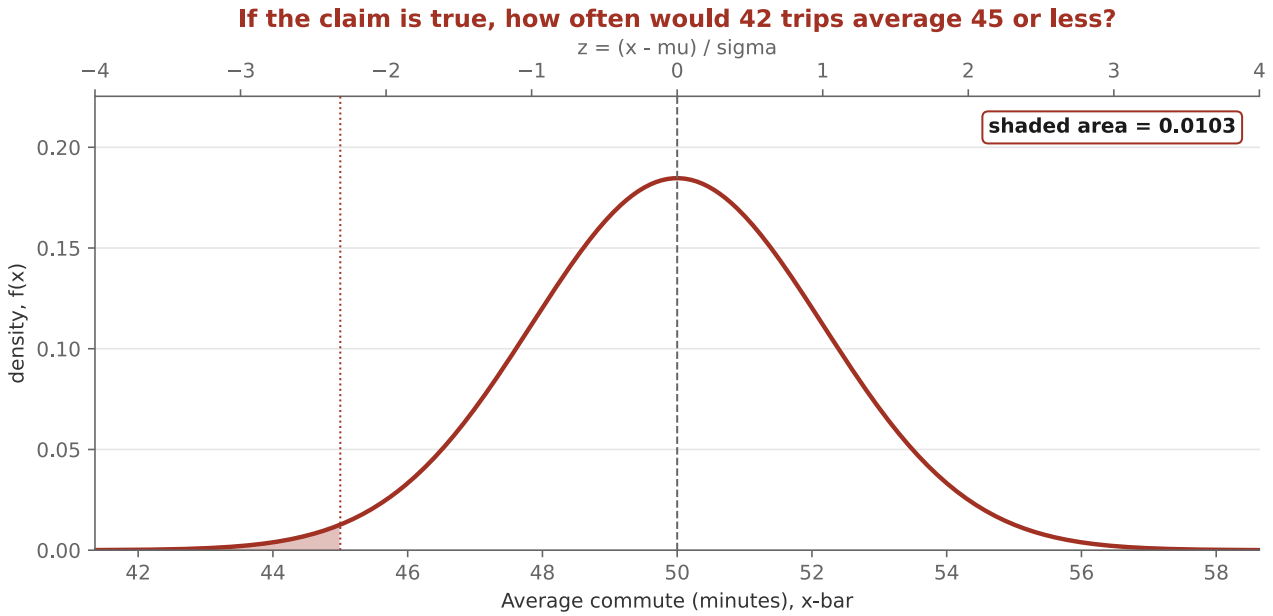
Using a Sampling Distribution to Judge a Claim

Back to the commute. You time 42 trips and they average **45 minutes**. Is the app's claim of 50 minutes believable?

The argument is the one from Chapters 5 and 6: **assume the claim, then ask how surprising the data would be.**

	45-minute sample average
standard error	2.160247

	45-minute sample average
finite population correction	1.0
event (x-bar)	X < 45
shape	left tail
z lower	-inf
z upper	−2.31455
P(X <= upper)	0.010319
P(X <= lower)	0.0
computed as	P(X <= 45)
probability	0.010319
Excel	=NORM.DIST(45, 50, 2.16025, TRUE)



If the true mean were 50 minutes, a sample of 42 trips would average 45 minutes or less only 1.03% of the time, about once in 97 samples. Either we drew a very unusual sample, or the claim is wrong. The sensible conclusion is that the average commute is probably *less* than 50 minutes. The 45 is also outside the 95% interval, 45.8 to 54.2, which is the same conclusion read a different way.

Mind the words. Had the sample averaged 48 minutes, inside the 95% interval, the right conclusion would be that the data are **consistent with** the claim, or **fail to contradict** it. Not that they "support" or "prove" it. A sample mean of 48 is also consistent with a true mean of 47 or 49; the data cannot single out 50. Your textbook says "support the claim" in several places. In Chapter 9 the distinction becomes the difference between "reject" and "fail to reject," and it is easier to learn it right now than to unlearn it then.

Kellogg's says boxes of Frosted Flakes average 18 ounces. A sample of 30 boxes averages 18.2 ounces. The filling process has a standard deviation of 0.32 ounces.

(a) If the claim is true, what is the interval that contains 95% of sample means for $n = 30$?

(b) How likely is a sample mean of 18.2 or more if the claim is true?

(c) Is the filling process working as Kellogg's says? If not, in which direction, and who should care?

i Solution

```
level              0.9500
z                  1.9600
standard error     0.0584
margin z x SE      0.1145
lower             17.8855
upper             18.1145
```

	x-bar >= 18.2
standard error	0.058424
finite population correction	1.0
event (x-bar)	X > 18.2
shape	right tail
z lower	3.423266
z upper	+inf
P(X <= upper)	1.0
P(X <= lower)	0.999691
computed as	1 - P(X <= 18.2)
probability	0.000309
Excel	=1 - NORM.DIST(18.2, 18, 0.0584237, TRUE)

(a) 17.885 to 18.115 ounces.

(b) 0.0003, about once in 3,232 samples.

(c) No: 18.2 is far outside the interval. The process is over-filling. Customers are not complaining; Kellogg's cost accountants should be, since every box carries about 0.2 ounces of free cereal. A claim can be wrong in a direction that only one party cares about.

Sampling from a Finite Population

So far the population has been effectively infinite: Nielsen's 41,000 households are a sliver of 129 million. When the sample is a sizeable share of the population, the standard error formula needs an adjustment.

When sampling **without replacement** from a population of size N, the standard error of the mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}.$$

> The factor $\sqrt{(N-n)/(N-1)}$ is the **finite population correction (FPC)**. Use it when the sample is more than 5% of the population, $n/N > 0.05$.

You have seen this factor before. In Chapter 5 it was the "shrink factor" that made the hypergeometric distribution less spread out than the binomial: every member you draw is one you cannot draw again, so a sample without replacement from a small population tells you more than the same-sized sample from a large one.

Why the correction exists

A pool-service company in Henderson has 100 residential customers. Their service ratings have historically averaged 7.2 on a 10-point scale, with a standard deviation of 0.7. Here is the standard error for several sample sizes, with and without the correction:

n	n / N	sigma / sqrt(n)	FPC sqrt((N-n)/(N-1))	corrected standard error
10	0.1	0.2214	0.9535	0.2111
40	0.4	0.1107	0.7785	0.0862
60	0.6	0.0904	0.6356	0.0574
80	0.8	0.0783	0.4495	0.0352
100	1.0	0.0700	0.0000	0.0000

Look at the last row. A sample of all 100 customers is a census: its mean *is* μ, and its sampling error is zero. The uncorrected formula still reports a standard error of 0.07. The correction brings it to exactly zero, and scales it down smoothly in between. Without it, you would overstate the uncertainty in a large sample from a small population.

A check by simulation: drawing 10,000 samples of 40 *without replacement* from a made-up population of 100 ratings with $\sigma = 0.7$, the sample means have a standard deviation of 0.0859. The corrected formula says 0.0862; the uncorrected one says 0.1107.

And when the population is large, the correction disappears on its own. For $n = 30$ from $N = 10,000$ it is 0.9985, which is why it can be dropped below the 5% threshold.

Application: has service improved?

A recent survey of 40 of the 100 customers averaged 7.5. If the true mean is still 7.2, how surprising is that?

	40 of 100 customers
standard error	0.086164
finite population correction	0.778499
event (x-bar)	X > 7.5
shape	right tail
z lower	3.481731
z upper	+inf
P(X <= upper)	1.0
P(X <= lower)	0.999751
computed as	1 - P(X <= 7.5)

	40 of 100 customers
probability	0.000249
Excel	=1 - NORM.DIST(7.5, 7.2, 0.086164, TRUE)

A sample average of 7.5 or more would happen about 0.02% of the time if nothing had changed. That is strong evidence that ratings across the customer base have risen. (Without the correction, the probability would be 0.34%; both say “rare,” but the corrected one is the right number.) The ordinal-data caution from the reviews example applies here too.

The Sampling Distribution of the Proportion

Many business questions are about a **share**: the share of households that watched, of graduates who found jobs, of orders that shipped late.

The **sample proportion** is

$$\hat{p} = \frac{x}{n},$$

where x is the number of successes in a sample of n. For samples from a population with proportion p, the sampling distribution of $\hat{p}$ has

$$\mu_{\hat{p}} = p \qquad \text{and} \qquad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}},$$

where $\sigma_{\hat{p}}$ is the **standard error of the proportion**. It is approximately normal when $np \geq 5$ and $n(1-p) \geq 5$, the same conditions as in Chapter 6, and

$$z = \frac{\hat{p} - p}{\sigma_{\hat{p}}}.$$

For a finite population with $n/N > 0.05$, multiply $\sigma_{\hat{p}}$ by the finite population correction.

This is not a new idea. The number of successes x is binomial, and Chapter 6 showed it is approximately normal with mean np and standard deviation $\sqrt{np(1-p)}$. Divide everything by n and you have the formulas above. A proportion is a count on a different scale. (Your textbook writes the sample proportion as $\bar{p}$; these notes use $\hat{p}$, which is what Chapter 8 and most software use.)

Application: Super Bowl viewership

A network sold advertising on the claim that 45% of US households would watch the Super Bowl. An advertiser surveys 200 households afterwards and finds 84 watched, $\hat{p} = 0.42$. How surprising is 84 or fewer, if the claim is true?

```
np = 90,  n(1 - p) = 110:  both at least 5, so the normal approximation is licensed.
standard error = sqrt(0.45 x 0.55 / 200) = 0.0352
```

	84 or fewer of 200
question	at most 84 of 200
p-hat at x	0.42
standard error	0.035178
z, uncorrected	−0.852803

	84 or fewer of 200
exact (binomial)	0.217495
normal, with continuity correction	0.217185
normal, no correction	0.196884

The continuity correction still matters. Your textbook drops it in this chapter, saying samples are large enough for it not to matter. For this very example it matters: the exact binomial answer is 0.2175, the normal approximation *with* the correction is 0.2172, and without it 0.1969, two percentage points off.

The policy in these notes:

- When the question is about a **count** ("84 or fewer households," "more than 58 buyers"), apply the continuity correction exactly as in Chapter 6. On the proportion scale, half a count is $0.5/n$.
- When the question is about a **proportion or percentage** ("less than 40% of the sample"), use the normal curve directly.
- When software can compute the exact binomial, as it always can, that is the number to report. The approximation is for understanding.

Now interpret. If 45% of households really watched, a sample showing 42% or less would turn up about 22% of the time, roughly one sample in five. That is not surprising. The sample **fails to contradict** the network's claim.

That is all it does. Your textbook says this result "supports" the claim, and it does not: 42% is also perfectly consistent with a true share of 40% or 43%. A large tail probability means the data have not given you a reason to doubt the claim. It does not mean they have given you a reason to believe it.

Application: a finite population of graduates

A college with 770 graduates claims 70% landed jobs in their field. A random survey of 120 graduates finds 97 did, $\hat{p} = 0.808$. The sample is 15.6% of the population, above 5%, so the finite population correction applies.

	97 or more of 120, N = 770
question	at least 97 of 120
p-hat at x	0.808333
standard error	0.03846
z, uncorrected	2.816758
exact (hypergeometric (N = 770, R = 539))	0.002625
normal, with continuity correction	0.00338
normal, no correction	0.002426

The exact answer, which for sampling without replacement from 770 graduates is hypergeometric, is 0.0026: a result this good would almost never happen if the true rate were 70%. The college is, if anything, understating its placement rate. All three methods agree on "very rare," though in a tail this far out they differ in the details; that is a reason to report the exact number.

Exercise

In a 2016 survey, 43% of US adults said they had looked at maps on their phones while driving. In a random sample of 125 adults:

(a) What are the mean and standard error of $\hat{p}$?

(b) What is the probability that 46 or more say they do?

(c) What is the probability that less than 40% of the sample say they do?

Solution

```
(a) mean = 0.43, standard error = sqrt(0.43 x 0.57 / 125) = 0.0443
```

	(b) 46 or more
question	at least 46 of 125
p-hat at x	0.368
standard error	0.044281
z, uncorrected	−1.400152
exact (binomial)	0.932806
normal, with continuity correction	0.931951
normal, no correction	0.919266

```
(c) P(p-hat < 0.40) = NORM.DIST(0.40, 0.43, 0.0443, TRUE) = 0.2490
```

(b) is about a count, so it gets the continuity correction: 46 or more starts at 45.5, which is $45.5/125$ on the proportion scale. **(c)** is about a percentage, so it uses the normal curve directly.

The Bootstrap: A Standard Error from One Sample

Every calculation so far needed σ or p, the population's own parameters. In practice you rarely know them. The **bootstrap** estimates a standard error from the one sample you have.

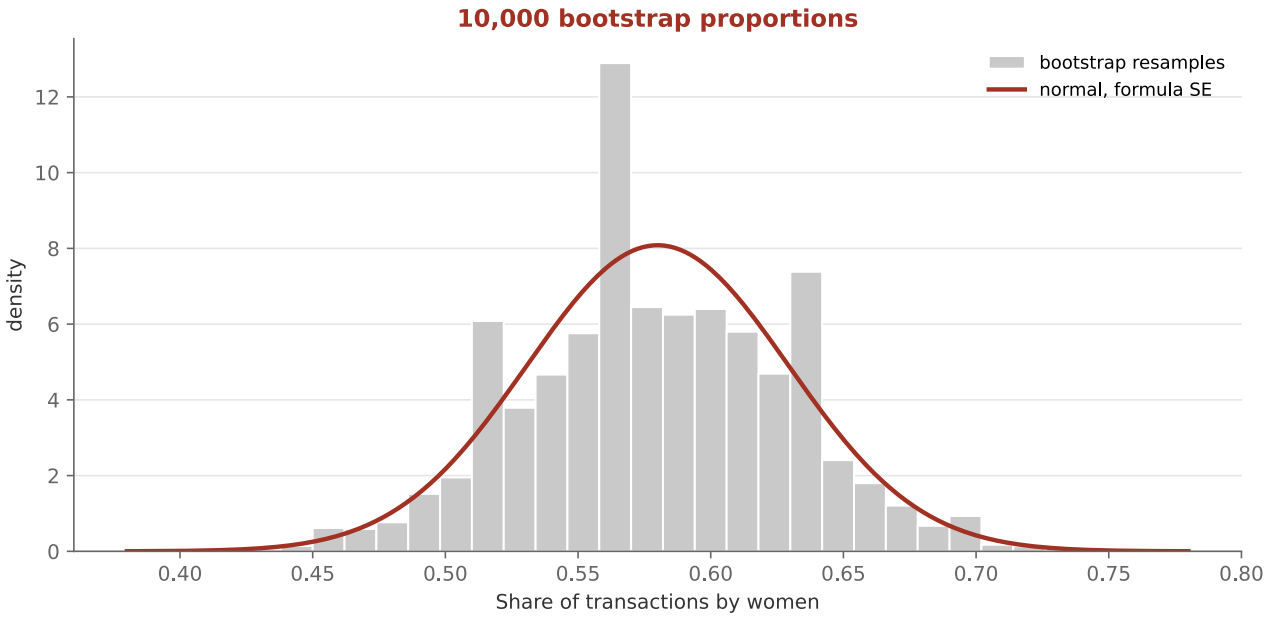
The **bootstrap** treats the sample as a stand-in for the population. It draws many **resamples** of the same size *from the sample*, with replacement, computes the statistic for each, and uses the spread of those statistics to estimate the standard error.

The bootstrap is not a way of collecting data, and it is not a fifth kind of probability sample, although your textbook lists it beside the sampling designs. It is a way of **analysing** a sample you already have. Every resample comes from that one sample, never from the population.

Application: female shoppers

A store manager examines a random sample of 100 transactions and finds 58 were made by women. How precise is $\hat{p} = 0.58$?

```
bootstrap standard error = 0.0494
formula sqrt(p-hat (1 - p-hat) / n) = 0.0494
middle 95% of bootstrap proportions: 0.480 to 0.670
```



The bootstrap standard error matches the formula, with no formula needed. That is the point: the same resampling works for a median, a ratio, a 90th percentile, or any statistic that has no tidy standard error formula. The middle 95% of the bootstrap proportions is a first look at the confidence interval of Chapter 8.

Application: faculty salaries

Take one simple random sample of 30 faculty, and pretend it is all we have.

	one sample of 30
sample mean	105090.0
bootstrap standard error	5859.0
s / sqrt(n), from the sample	5946.0
true standard error, sigma / sqrt(n) x FPC	4993.0
bootstrap 95%: lower	93621.0
bootstrap 95%: upper	116708.0
true population mean	93457.0

From one sample of 30, the bootstrap recovers a standard error in the right neighbourhood of the true one, $5,859 against $4,993, which we can check only because we happen to have the whole population. Notice, though, that this particular sample drew high: its mean is $105,090, and its 95% interval just misses the true mean of $93,457. That is not a bug. About one sample in twenty will do this, and Chapter 8 is about living with that fact.

Which Formula Is It?

The question is about	Center	Standard error	Correction if $n/N > 0.05$
one individual value x	μ	σ (not a standard error)	none

The question is about	Center	Standard error	Correction if $n/N > 0.05$
a sample mean $\bar{x}$	μ	$\sigma/\sqrt{n}$	$\times \sqrt{(N-n)/(N-1)}$
a sample proportion $\hat{p}$	p	$\sqrt{p(1-p)/n}$	$\times \sqrt{(N-n)/(N-1)}$
a count of successes x	np	$\sqrt{np(1-p)}$	continuity correction ± 0.5

For each, name the row of the table and the standard error.

(a) Hotel room rates average \$155 with a standard deviation of \$26. What is the probability that a randomly chosen hotel charges more than \$163?

(b) Same rates. What is the probability that the average of 35 randomly chosen hotels exceeds \$163?

(c) 58% of Generation Y pay their bills on time. In a sample of 200, what is the probability that 125 or fewer do?

(d) A college has 240 employees on its health plan, with mean out-of-pocket cost \$1,880 and standard deviation \$515. An audit samples 30. What is the probability the sample mean is below \$1,825?

i Solution

	row	standard error	probability
a	one value	26.0000	0.3792
b	sample mean	4.3948	0.0344
c	count	6.9800	0.9139
d	sample mean, finite	88.1368	0.2663

(a) and **(b)** ask about the same \$163, and the answers differ by a factor of more than ten. **(a)** is also the one question here that the CLT cannot answer unless room rates themselves are normal. **(c)** is a count, so the exact binomial is reported (the corrected normal gives nearly the same). **(d)** samples 30 of 240, which is 12.5%, so the finite population correction applies.

Laboratory 7

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 7 pack — one file. Unzip it and you will have a folder holding 8 files. Drag all 8 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 8 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `07_sampling_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 7 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.
>
> Then run `07_sampling_distributions.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach the page, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Your own junior analyst

The lab script answers the questions in this note. The skill to practice is getting a *new* question answered and checking the answer. Try this, with your own numbers in place of mine:

PROMPT

> A casino's slot floor averages $42 of net win per machine-hour, with a standard deviation of $65. A manager will audit a random sample of 50 machine-hours. By running code, find (1) the standard error of the sample mean, (2) the probability that the sample averages less than $30, and (3) the interval that contains 95% of sample means. Before computing, tell me which standard deviation goes in the denominator of the z-score and why. Then tell me whether n = 50 is large enough for the Central Limit Theorem here, given what net win per machine-hour probably looks like, and how you would check.

Check three things in the reply: that the z-score divides by $65/\sqrt{50}$ and not by 65, that the numbers came from executed code, and that the answer to the last question is not simply "yes, because 50 is more than 30." Net win is skewed and can be negative; a good answer says so and proposes a simulation like the Nevada AGI one.

Working in Excel

PROMPT

> Build me an Excel workbook that demonstrates the Central Limit Theorem. Sheet 1 holds a skewed population of 2,000 values (use =-LN(RAND())*4 so they are exponential with mean 4). Sheet 2 draws 500 samples of size n, where n is an editable input cell, using INDEX and RANDBETWEEN, and computes each sample's mean. Report the mean and standard deviation of the 500 sample means beside mu and sigma / SQRT(n), and draw histograms of the population and of the sample means with the vertical axis titled "count". Give me the file to download.

Set n to 2, then 10, then 30, pressing F9 a few times at each to redraw. Answer in one paragraph: at what n would *you* be comfortable treating the sample means as normal for this population, and what did you look at to decide?